

SOURAJ SHIL

DATA ENGINEERING

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PROFILE SUMMARY

Computer Applications undergraduate focused on Data Engineering, with hands-on experience building end-to-end ETL/ELT pipelines using Python, SQL, PySpark, Apache Airflow, dbt, PostgreSQL, and AWS. Built cloud and local production-style pipelines covering ingestion, data quality, incremental processing, data warehousing, orchestration, and analytics.

PROJECTS

Stock Data Pipeline

Python • Pandas • yfinance • PostgreSQL • Streamlit • APScheduler

- Built a modular stock-data pipeline using Python, Pandas, and yfinance for market-data ingestion.
- Structured fetch, transform, PostgreSQL load, Streamlit dashboard, and APScheduler components.
- Prepared the pipeline for automated execution and interactive stock-data visualization.

End-to-End E-Commerce Data Pipeline

Airflow • dbt • PostgreSQL • Docker • Metabase • Python • Faker

- Built a production-style daily ETL/ELT workflow: Python/Faker → PostgreSQL → dbt staging/intermediate/marts → Metabase.
- Implemented incremental models, idempotent writes, watermarking, data-quality checks, dbt tests, and retry/logging patterns.
- Created daily-sales, product-performance, and customer-RFM analytical data marts.

Books Data Pipeline

Airflow • PySpark • S3 • Redshift • Terraform • Web Scraping

- Built an hourly pipeline: scraping → raw JSON in S3 → PySpark cleaning/deduplication → partitioned Parquet → Redshift upsert.
- Provisioned AWS infrastructure with Terraform and added Airflow orchestration, analytics views, and monitoring.

Superstore Sales Data Engineering & BI Analytics Dashboard

Microsoft Fabric • Lakehouse • PySpark • Python • SQL • Power BI

- Ingested and stored Superstore sales data in Microsoft Fabric Lakehouse; authored PySpark/Python/SQL cleaning notebooks.
- Created cleaned gold layer dataset and built interactive Power BI dashboard with sales KPIs, regional growth, and product trends.

E-Commerce Product Recommendation System

Python • Pandas • Scikit-Surprise • Streamlit • TF-IDF

- Built a hybrid recommendation model combining SVD matrix factorization collaborative filtering and TF-IDF cosine similarity.
 - Evaluated model using Precision@5, Recall@5, and NDCG@5; deployed interactive discovery interface on Streamlit.
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EDUCATION

Bachelor of Computer Applications (BCA) — Assam Down Town University, 2024–2027 (Pursuing)
12th (CBSE) — 67% | 2024 • **10th (CBSE)** — 55.6% | 2022

TECHNICAL SKILLS

Languages: Python, SQL, C

Data Engineering: ETL/ELT, Data Quality & Gating, Incremental Loads, Data Warehousing, Star Schema Modeling, Idempotent Writes

Big Data & Cloud: Apache Spark, PySpark, Parquet Optimization, Amazon S3, AWS Glue, AWS Lambda, Amazon Redshift, EventBridge, IAM, Secrets Manager, CloudWatch, Microsoft Fabric

Orchestration, BI & Tools: Apache Airflow, dbt, Docker, Terraform, PostgreSQL, Power BI, Metabase, QuickSight, Pandas, Streamlit, Git, GitHub

CERTIFICATIONS

C for Everyone: Programming Fundamentals — Coursera • Introduction to Git & GitHub • Introduction to Data Engineering (IBM) • Introduction to Data Science (IBM) • Introduction to Artificial Intelligence (AI) • Introduction to Contemporary Operating Systems